Geoderma 368 (2020) 114313

Contents lists available at ScienceDirect

Geoderma

journal homepage: www.elsevier.com/locate/geoderma

The interplay between abiotic factors and below-ground biological interactions regulates carbon exports from peatlands

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ARTICLE INFO

Handling Editor: Daniel Said-Pullicino

Keywords:
Dissolved organic carbon
Plant biomass
Soil mesofauna
Soil respiration
Sphagnum
Vascular plants

ABSTRACT

Climate change projections indicate increases in CO_2 emissions and DOC release from peatlands, thereby potentially transforming them from sinks to carbon sources. The individual effects of abiotic and biotic factors on peatland carbon exports and the interactions among all these factors are still poorly understood. Therefore, we conducted a field study over two years in a peatland complex by selecting four habitats differing in their dominant vegetation (shrub, grass, sedge and moss) in order to assess the links between abiotic and biotic factors and carbon release at each habitat. Accounting for the wide variability across habitats allowed us to examine the direct and indirect effects of abiotic and biotic factors on carbon cycling at the ecosystem level (i.e. the whole peatland complex). Results showed that the habitats dominated by vascular plants had the highest microbial and above-ground plant biomass, as well as the highest values of Acari and Collembola biomass, whereas Enchytraeidae biomass was dominant in the moss habitat. Furthermore, at ecosystem level, the SEM explained 75% of the total soil respiration variation and 33% of the DOC variation. Accordingly, soil respiration was directly linked to enchytraeid biomass, with these invertebrates being positively related to soil moisture. In contrast, both soil temperature and soil moisture had direct effects on microbial and Acari biomass, and the interaction between Acari biomass and below-ground plant biomass was responsible for DOC release. These findings provide new insights into the role of soil biota in carbon dynamics in peatlands, by showing that the direct effects of abiotic factors on biotic communities (both plants and soil organisms) determine how much carbon is lost from these systems (either as CO_2 e or DOC).

1. Introduction

Peatlands only cover 3% of the global terrestrial surface, but they store 15–30% of the world's soil carbon (Limpens et al., 2008). Waterlogging conditions, acid pH and low nutrient availability determine their low decomposition and high carbon accumulation rates (Gorham, 1991). The main sources of carbon exports from peatlands are CO_2 and CH_4 emissions and the leaching of dissolved organic carbon (DOC), which are intrinsically linked to the dominant above-ground vegetation that modulates the soil environment (Dorrepaal, 2007; Ward et al., 2013; Robroek et al., 2015; Gavazov et al., 2018). DOC release can range between 1 and 50 g C m^{-2} yr^{-1}, whereas CO_2 exchange ranges between 20 and 60 g C m^{-2} yr^{-1}, In contrast, CH_4 exchange has a lower contribution with ranges between 5 and 8 g C m^{-2} yr^{-1} (Limpens et al., 2008).

Linking above- and below-ground communities is essential for a better understanding of biogeochemical cycles (Wardle et al., 2004). Plants, soil fauna and microbes interactions affect the turnover of soil carbon (Bardgett and Van der Putten, 2014). Specifically, plant-soil interactions occurring in the so-called rhizosphere – the plant root-soil interface inhabited by unique communities of soil biota - play an important role in carbon sequestration and exchange (Bonkowski et al., 2009; Huang et al., 2014; Finzi et al., 2015). Dominant vegetation with different growth patterns provides a wide variety of food sources with varying quality for below-ground decomposers, thereby affecting carbon turnover (De Deyn et al., 2008; Chen et al., 2016). For example, in peatlands, mosses, ericaceous shrubs and sedges produce litter with different lignin and polyphenol contents that result in dissimilar effects on soil respiration and organic matter decomposition rates (Ward et al., 2009). Some climate change projections have anticipated a functional shift in peatlands plant communities that will favour vascular plants (especially ericaceous shrubs) against mosses (Gunnarsson et al., 2002;

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https://doi.org/10.1016/j.geoderma.2020.114313
Received 25 October 2019; Received in revised form 29 January 2020; Accepted 3 March 2020
Available online 13 March 2020
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Weltzin et al., 2003). Consequently, this change is expected to alter below-ground communities and, probably, the sink function of these ecosystems (Bragazza et al., 2013; Dieleman et al., 2017).

The individual effects of abiotic and biotic factors on carbon turnover in peatlands have been examined in several studies (e.g. Silvola et al., 1996; Laine et al., 2012). Nevertheless, the results are inconsistent and controversy remains on which are the main drivers of greenhouse gas emissions and DOC release. Studying abiotic-biotic interactions provides a more holistic view of ecosystem processes (Bardgett, 2005). Accordingly, some studies have analysed the potential interactions between abiotic and biotic factors and their effects on peatland carbon exports, and indicated that both plant and soil communities are modulated by environmental variables and contribute to carbon turnover (e.g. Cole et al., 2002; Carrera et al., 2011; Briones et al., 2014; Armstrong et al., 2015). However, certain soil biota activities, such as feeding and burrowing, are not usually taken into account when studying soil organic matter dynamics, despite the fact that incorporating soil fauna in soil models can drastically change the predictions of carbon cycle (Filser et al., 2016). For instance, microarthropods can exert a top-down control of microbial communities (Grandy et al., 2016; Soong and Nielsen, 2016) and enchytraeids have shown to increase carbon turnover (Cole et al., 2002; Carrera et al., 2009). However, only a few studies have included soil fauna as an input variable in carbon models (De Vries et al., 2012a, 2013; Tresch et al., 2019).

In this study, we investigated how the interplay between abiotic and biotic factors could affect carbon release (CO_2 and DOC) in a peatland complex by selecting four habitats that accounted for a wide range of abiotic and biotic conditions representative of the whole ecosystem. The four selected habitats differed in their dominant vegetation (shrub, grass, sedge and moss). We took plant and soil samples every two months during two years to test the following hypotheses: 1) carbon turnover and soil biological properties differ between habitats dominated by vascular plants (shrub, grass and sedge) and non-vascular plants (moss); 2) at the ecosystem level, abiotic factors indirectly influence carbon exports through their direct effects on soil biological properties, and 3) dominant soil biota (i.e. microorganisms and mesofauna) influence carbon release in peatlands. By obtaining detailed spatial and temporal information about abiotic (soil temperature, soil moisture and soil chemical properties) and biotic variables (above- and below-ground plant, microbial and mesofaunal biomass) and their links to soil respiration and DOC production we could quantify the effects of the abiotic-biotic interactions on carbon dynamics.

2. Materials and methods

2.1. Study area

We conducted the study at Serra do Xistral (Lugo, NW Spain), a mountain range classified as a Site of Community Importance (code ES1120015) and located in the southernmost part of the Atlantic Biogeographic Region. The whole study area comprises various highly differentiated natural habitat types that contribute to the maintenance of biological diversity at a favourable state according to the Habitats Directive (92/43/EEC). Specifically, we selected four peatland habitats where only one or a few plant species are dominant: (i) an active blanket bog dominated by the sedges *Eriophorum angustifolium* and *Carex durieui* (43° 30′ 12″ N, 7° 33′ 02″ W; 970 m a.s.l.; code 7130*), (ii) an active blanket bog with the purple moor-grass *Molinia caerulea* dominating the plant community (43° 27′ 36″ N, 7° 34′ 12″ W; 960 m a.s.l.; code 7130*), (iii) an Atlantic wet heath dominated by *Erica mackayana* and *Calluna vulgaris* (code 4020*), and (iv) a transition mire consisting of a moss carpet of *Sphagnum* spp. (code 7140*). These last two mentioned habitats are both located in the same valley area (43° 26′ 56″ N, 7° 33′ 61″ W; 714 m a.s.l.). The peat of the moss habitat had the highest total porosity (94%) and total fibre content (49%) when compared to the other three habitats (shrub: 81.9% and 6.7%, grass: 5.3% and 86.2%, and sedge: 86.8% and 24.2%, respectively; Juan-Ovejero et al., 2019). The climate of the area is oceanic, with a mean annual temperature of 10.6 °C (being February the coldest month with 6.0 °C and August the warmest one with 16.0 °C) and annual rainfall of 1533 mm (www.meteogalicia.gal). Foggy conditions are frequent during the whole year, and snow and frosts normally occur in cold months.

2.2. Sample collection and measurements

The field study took place from January 2016 to November 2017. At each of the four peatland habitats, we recorded soil temperature and soil respiration and collected plant and soil samples for determinations of biological communities and soil properties.

2.2.1. Soil temperature records

At each habitat we inserted a temperature data logger (UA-002–08 HOBO) 5 cm into the soil in order to register soil temperature every hour during the whole study period. Due to temporal interruptions in the data loggers, some soil temperature values were missed and estimated by triangulating temperature data from the three nearest meteorological stations (Lanzós: 43° 22′ 28.56″ N, 7° 38′ 40.85″W; Fragavella: 43° 27′ 16.56″ N, 7° 26′ 46.5″W and Borreiros: 43° 37′ 51.24″ N, 7° 37′ 51.17″ W; www.meteogalicia.gal). We used machine learning algorithms in WEKA parameters (Waikato Environment for Knowledge Analysis; Frank et al., 2016) to obtain the missing values (for full details of the extrapolation procedure see Juan-Ovejero et al., 2019).

2.2.2. Soil respiration measurements

At each habitat, we inserted five PVC cylindrical collars (10 cm diameter × 10 cm depth) into the soil on the first sampling occasion (January 2016), which remained in place for the entire investigated period. We did this to avoid an overestimation of the soil CO_2 efflux associated with perturbations due to the insertion of the PVC collars (Heinemeyer and McNamara, 2011; Jovani-Sancho et al., 2017). We measured soil respiration every two months (since March 2016) using a portable infrared gas analyser LI-COR 8100A attached to a survey chamber of 10 cm diameter (LI-COR, Lincoln, Nebraska). Data were expressed as mg C $m^{-2}h^{-1}$.

In order to examine the temperature sensitivity of soil respiration we also calculated the Q_{10} quotient according to Luo et al. (2001), which is based on the exponential function $Q_{10} = e^{10b}$. We used a mixed model to calculate the Q_{10} quotient by considering the log-transformed soil respiration data as a function of the soil temperature (i.e. fixed factor) and using the sampling date as a random effect. We applied the following equation using the output of the mixed model:

$$\ln(R_s) = a + bT$$

where R_s is the soil respiration on each sampling occasion, T is the average soil temperature of the 3 days before sampling (Bragazza et al., 2016), a the intercept of R_s when T is zero, and b the temperature sensitivity coefficient. Finally, we calculated the Q_{10} quotient using the primary exponential function.

2.2.3. Plant materials

Three intact soil cores (10 cm diameter × 14 cm depth) were collected every two months at each habitat and stored at 4 °C until processing. Above- and below-ground (0–10 cm) plant materials were hand-sorted, with vegetation being gently washed from the soil with tap water (Pérez-Harguindeguy et al., 2013). Thereafter, all above- and below-ground materials were oven-dried at 65 °C for 48 h. The results were expressed as g m^{-2}.

2.2.4. Soil samples for determinations of biological and soil properties

At each of the four habitats, we randomly collected ten additional intact soil cores (10 cm diameter × 14 cm depth) every two months. From each one, we took three subsamples down to a depth of 10 cm: two for mesofaunal extractions (microarthropods: approximately 40 cm^2 and enchytraeids: 20 cm^2) and one for soil moisture determinations (20 cm^2). Samples were placed in plastic bags and stored at 4 °C until processing a few days later. We extracted microarthropods (mites and collembolans) using the Tullgren method (Tullgren, 1918) and enchytraeids with the wet extraction method (O'Connor, 1955). Abundances of all mesofauna groups were first expressed as individuals m^{-2}, but then converted to biomass values using Petersen (1982) estimates. Average dry weights per individual were 4.3 μg (Acari), 2.7 μg (Collembola) and 32 μg (Enchytraeidae). We assumed a carbon content for Acari and Collembola of 47.7% and 50% for enchytraeids (Berg et al., 2001), and the metabolically active carbon (i.e. carbon assimilated by animals) of each individual was considered to be ¾ of their total carbon content (Brown et al., 2004; Woodward et al., 2005). Following this, Acari, Collembola and Enchytraeidae biomass values were finally expressed as mg active C m^{-2}.

Soil moisture was measured gravimetrically by oven-drying the fresh soil at 105 °C for 48 h or until constant weight and re-weighing. Data were expressed as percentage.

An additional set of three intact soil cores of the same size as before was used for total C and N contents and microbial biomass determinations. Soil samples (5 g) were freeze-dried and sieved (< 2 mm) and total carbon and nitrogen contents were measured by means of a LECO elemental analyser (CN-2000, LECO Corp., St Joseph, MI). Results were expressed as C/N ratio.

Microbial biomass carbon (C_{mic}) was obtained using the fumigation-extraction method (Vance et al., 1987). Two root-free soil subsamples equivalent to 10 g (fresh weight) were taken from every soil core, with one being fumigated with 25 ml of ethanol-free $CHCl_3$ for 24 h, and with the other representing the non-fumigated subsample. Both subsamples were then extracted with 100 ml 0.5 M K_2SO_4, centrifuged with an orbital shaker (Selecta® Rotabit) for 30 min at 200 rpm, and the final solution filtered (FilterLab® No. 1252, 7–9 μm pore size). The extracts were then frozen until further analyses. Microbial biomass carbon was calculated as $C_{mic} = E_c/k_c$; where E_c is the difference between the organic C extracted from the fumigated and non-fumigated subsamples and k_c is the fraction of mineralized C, which takes the value of 0.45 in soils with pH ≥ 4.2 (Vance et al., 1987).

For DOC determinations, three additional intact soil cores of smaller size (5.5 cm diameter × 10.5 cm depth) were also collected at each habitat on each sampling occasion. Soil samples were leached by immersion in 200 ml of distilled water and draining under gravity (Anderson and Ineson, 1982). The leachates were filtered (FilterLab® No. 1252, 7–9 μm pore size) and frozen until analysis. Total dissolved organic C in the microbial extracts and leachates was measured with a Shimadzu Total Organic Carbon Analyser (TOC-5000A) equipped with an autosampler ASI-V. Microbial biomass was corrected for soil dry weight and DOC was corrected for water volume. The results were expressed as mg C m^{-2}.

The pH of the soil solutions was also measured using a Crison micropH 2000 and combination electrode.

2.3. Statistical analyses

All analyses were performed with R 3.5.2 (R Core Team, 2018). Data were inspected for normality and homoscedasticity of variance prior to analyses, and transformed when necessary. Accordingly, biomass of the three mesofauna groups (mites, collembolans and enchytraeids) were transformed to $\log_{10}(x + 1)$, microbial biomass to square-root(x), soil moisture to arcsin(x), C/N ratio, above- and below-ground plant biomass, DOC and soil respiration were log-transformed, whereas soil temperature and pH of the leachates remained untransformed.

Firstly, we tested for significant differences between habitats by performing linear mixed-effects models. Specifically, we applied independent linear mixed-effects models with C/N ratio, pH of the leachates, soil moisture, above- and below-ground plant biomass, Acari biomass, Collembola biomass, Enchytraeidae biomass, microbial biomass and DOC as fixed factors and sampling date as a random factor and used a restricted maximum likelihood procedure. In addition, we ran independent repeated measures linear mixed-effect models with soil temperature and soil respiration as fixed factors and sampling date as a random factor and used a restricted maximum likelihood procedure. A post hoc Tukey HSD test ($p < 0.05$) was used for pairwise comparisons between habitats whenever the models indicated that significant differences were present. Models were obtained using the 'nlme' package (Pinheiro et al., 2018).

Secondly, we analysed both the effects of habitat, abiotic and biotic factors (i.e. explanatory variables) on carbon release (i.e. response variables) using model selection through Generalized Linear Mixed Models (GLMM) of the gamma family with log as link function, abiotic and biotic variables as fixed factors and sampling date as a random factor. Habitat effect was not statistically significant and it was removed from further analyses. We then analysed our well-replicated dataset to investigate the interactive effects between all factors on carbon exports at ecosystem level. In order to select the best fitting models, we used the Type II Sum of Squares for variable addition, and the Chi-Square Wald test ($p < 0.05$) served to determine which models resulting from different combinations of explanatory variables and factor interactions were significant. We used the Penalized Quasi-likelihood (PQL) approach to determine asymptotic bias in the regression coefficients and variance components ('MASS' package; Venables and Ripley, 2002). Because biotic factors and response variables showed different ranges of variation, they needed to be standardized prior to model fitting. In addition, a positive integer (+1) was added to invertebrate biomass values to avoid negative values.

Spearman correlation analysis was used to check for collinearity prior to model selection. As result, above-ground plant biomass had to be removed from the model since it was partially correlated with below-ground plant biomass. We plotted the residuals and residuals *vs.* fitted values to validate the model assumptions. R^2 of the models was calculated with the 'r2glmm' package (Jaeger, 2017).

Although variable addition in linear models does not allow us to unravel causality between biotic communities, soil properties and carbon stocks (Wardle et al., 2004; de Vries et al., 2012b), Structural Equation Modelling (SEM) has proven to be a valuable statistical tool to better understand ecological networks (Shipley, 2009; Eisenhauer et al., 2015). Therefore, we performed Piecewise Structural Equation Modelling (piecewise SEM within the 'piecewiseSEM' package; Lefcheck, 2016) to detect any direct and indirect relationships among all explanatory variables (i.e. with abiotic and biotic factors as input variables) that could influence the response variables (here, soil respiration and DOC or output variables) at the ecosystem level. SEM needs to postulate an *a priori* model that proposes which relationships can be plausible (Grace, 2006). Therefore, we hypothesized that soil moisture and soil temperature would indirectly affect carbon release (soil respiration and DOC) through their direct effects on soil biological properties (i.e. below-ground plant, microbial and mesofaunal biomass). We used General Linear Mixed Models (GLMMs) with our previously selected abiotic and biotic factors as fixed factors and sampling date as a random factor, and included any potential interactions between soil organisms (e.g. trophic interactions) and the relationships between soil respiration and DOC (which might not be unidirectional) as correlated errors (Shipley, 2009; Lefcheck, 2016). The total sample size we used to perform piecewise SEM was 4622. Significant relationships ($p < 0.05$) between abiotic and biotic factors, among biotic factors, and between these factors and C release were included in the final piecewise SEM. We also provide the Akaike's Information

Table 1
Mean values (with 95% confidence intervals) of the abiotic factors measured at each investigated habitat. Soil temperature (average of monthly values), soil moisture (n = 10 per habitat and sampling date), pH (n = 3 per habitat and sampling date) and C/N ratio (n = 3 per habitat and sampling date). Different letters indicate differences between habitats ($p < 0.05$).

	Shrub (95% C.I.)	Grass (95% C.I.)	Sedge (95% C.I.)	Moss (95% C.I.)
Soil temperature (°C)	11.24 (8.47–14.01) a	9.88 (7.35–12.41) b	10.10 (7.66–12.54) b	11.96 (9.14–14.78) a
Actual soil moisture content (%)	68.70 (67.42–69.98) d	77.03 (75.73–78.34) c	82.64 (82.17–83.11) b	86.98 (86.74–87.22) a
pH	5.94 (5.84–6.04) c	6.13 (6.03–6.24) ab	6.28 (6.18–6.38) a	6.06 (5.98–6.15) bc
C/N ratio	19.33 (18.71–19.95) b	20.51 (19.56–21.46) b	22.12 (21.42–22.82) a	19.43 (18.84–20.02) b

Criterion (AIC) and Bayesian Information Criterion (BIC) that aim at facilitating model selection by comparing the fit of all produced models (Shipley, 2009, 2013). All investigated variables included in our *a priori* model were inter-related directly or indirectly because all hypothesis were considered plausible, and therefore, we could not perform the global goodness of fit with the Fisher's C statistic.

3. Results

3.1. Abiotic factors at each habitat

Abiotic factors differed between the four habitats (Table 1). Thus, soil temperature was higher in the habitats located at a lower elevation (i.e. habitats dominated by shrubs and mosses) than in those at higher locations (i.e. habitats dominated by sedges and grasses) (Table 1). Soil moisture showed the highest values at the moss habitat and the lowest ones at the shrub habitat (Table 1). The sedge habitat had the highest values of pH of the leachates (Table 1) and soil C/N ratio (Table 1), whereas the shrub habitat showed the minimum values of these two variables (Table 1).

3.2. Biotic factors and carbon exports at each habitat

Above-ground plant biomass was higher in the three habitats dominated by vascular plants than in the moss habitat (Fig. 1a). In addition, below-ground plant biomass was lower under the shrub than under the grass and moss habitats, whereas the sedge habitat showed an intermediate position (Fig. 1b). In contrast, microbial biomass (Fig. 1c) was significantly lower at the habitat dominated by mosses than in those dominated by vascular plants.

Overall, Enchytraeidae biomass (Fig. 1d) was higher than those of Acari (Fig. 1e) and Collembola (Fig. 1f). Mesofaunal biomass also differed significantly between habitats: Enchytraeidae biomass was significantly higher in the habitats dominated by grasses and mosses (Fig. 1d), whereas the highest biomass values of Acari (Fig. 1e) and Collembola (Fig. 1f) were observed at the grass habitat (although high biomass values of Acari were also measured at the shrub habitat).

Soil respiration was significantly higher in the habitat dominated by shrubs than in the moss and grass habitats, and the sedge habitat occupied an intermediate position (Fig. 2a). However, the highest Q_{10} value was measured at the grass habitat (5.68, C.I. (95%): 2.75–8.61), followed by the one with mosses (5.20, C.I. (95%): 3.03–7.25), the shrub habitat (4.23, C.I. (95%): 1.87–6.59), and finally, the lowest value corresponded to the sedge habitat (3.96, C.I. (95%): 1.75–5.68) (Fig. S1). In contrast, the highest DOC values were measured at the moss habitat and the lowest ones at the shrub habitat (Fig. 2b).

3.3. Modelling the effects of abiotic and biotic factors on carbon release at the ecosystem level

3.3.1. Preliminary model fitting

Table 2 shows the results from the model selection (pooled dataset) of the abiotic and biotic variables that played a more important role in carbon release at the investigated habitats. We firstly fitted the models using only abiotic factors and accordingly, we chose soil moisture and soil temperature as explanatory variables for soil respiration, and soil moisture for DOC, since they were significant in the model fitting. In addition, we also fitted the models using only biotic factors as explanatory variables. This resulted in three models with DOC as response variable: the first model had microbial biomass and Acari biomass as explanatory variables, the second one was explained by below-ground plant biomass and Acari biomass, and the third model had below-ground plant biomass, microbial biomass and Collembola biomass as explanatory variables. In the case of soil respiration, two significant models were selected, one had microbial biomass and Enchytraeidae biomass as explanatory variables, and the other was explained by below-ground plant biomass and Enchytraeidae biomass. For both response variables interactions between factors were not statistically significant.

3.3.2. Piecewise structural equation modelling

Piecewise SEM (Fig. 3) tested the direct and indirect effects of abiotic (soil temperature, soil moisture) and biotic factors (microbial biomass, below-ground plant biomass, Acari biomass, Collembola biomass and Enchytraeidae biomass) on soil respiration and DOC at the ecosystem level (AIC = 96.000, BIC = 185.818). The final model explained 75% of the total soil respiration variation and 33% of the DOC variation (Fig. 3). The results showed that high soil moisture contents had a direct negative effect on Collembola, Acari and microbial biomass, but a positive effect on Enchytraeidae biomass. Soil temperature had a direct negative relationship with Acari and microbial biomass. Below-ground plant biomass affected DOC indirectly through Acari biomass, whereas Enchytraeidae biomass had a direct effect on soil respiration (Fig. 3).

4. Discussion

Our study shows that biotic factors and carbon exports differed between peatland habitats, emphasizing the importance of the feedback mechanisms between plants and soil biota in the habitat context (i.e. in terms of plant community and abiotic factors; Bardgett et al., 2005).

In agreement with previous studies (e.g. Bonnett et al., 2006; Bell et al., 2018), our results indicated that differences in carbon release between habitats were modulated by abiotic factors (namely soil moisture and soil temperature). Accordingly, lower rates of soil respiration were measured in those habitats with higher soil water contents (sedge, grass and moss) than in the shrub habitat, which had the highest soil temperatures and the lowest soil moisture contents. This is coincidental with previous results indicating the existence of a positive relationship between peat aeration, high temperatures and CO_2 emissions (Silvola et al., 1996; Bubier et al., 2003). This is partly because anoxic conditions prevent the enzyme phenol oxidase from degrading phenolic compounds, thereby inhibiting carbon transformations (Freeman et al., 2004; Dunn and Freeman, 2018). Also, in agreement with previous studies (Gibson et al., 2009; Preston et al., 2011), DOC production was related to high water tables, and the wettest habitat (moss) exhibited the highest DOC release, whereas the driest one (shrub) showed the lowest value.

In contrast to the study by Bragazza et al. (2016) we did not find a clear pattern that could indicate lower Q_{10} quotients at lower soil

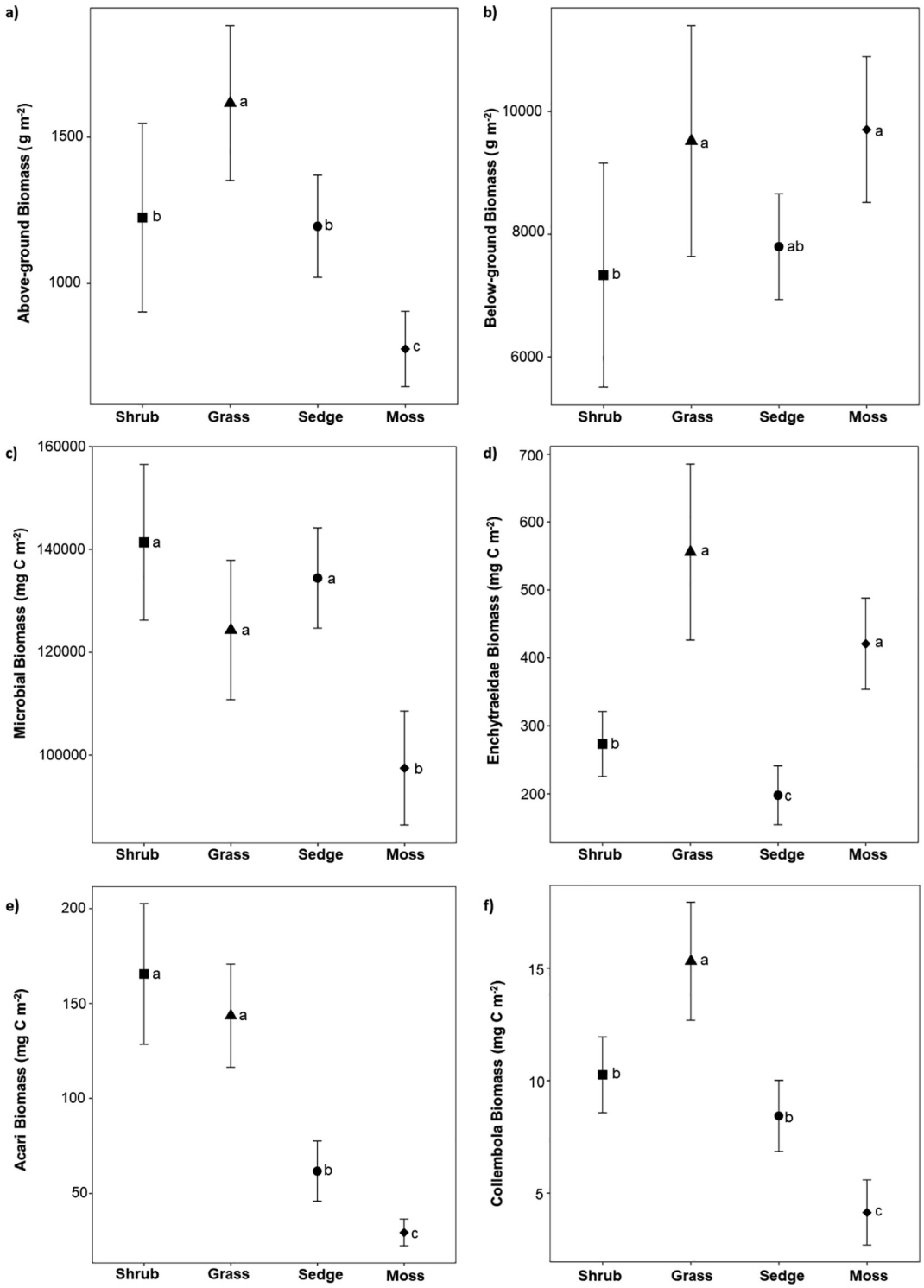


Fig. 1. Mean values (± 95% confidence intervals) of biotic factors: (a) above-ground plant biomass, (b) below-ground plant biomass, (c) microbial biomass, (d) Enchytraeidae biomass, (e) Acari biomass and (f) Collembola biomass (n = 3 per habitat and sampling date for above-, below-ground plant and microbial biomass; n = 10 per habitat and sampling date for Enchytraeidae, Acari and Collembola biomass). Different letters indicate differences between habitats ($p < 0.05$).

moistures. Nevertheless, the sedge habitat showed the highest C/N ratio and the lowest Q_{10} quotient, which is consistent with the fact that high carbon contents and low temperature sensitivity of soil respiration are linked to slow carbon turnover and long-term carbon accumulation (Briones et al., 2014).

In addition, the three habitats dominated by vascular plants showed higher values of microbial biomass than the moss habitat, which is in agreement with the results of Kaštovská et al. (2018) and it can be explained by the abundance of polyphenols that inhibit microbial activity (Bragazza and Freeman, 2007). However, the moss habitat contained a higher biomass of Enchytraeidae worms, and this result agrees with that of Carrera et al. (2011), who indicated that, under favourable moisture conditions, higher enchytraeids activities could enhance DOC release. Importantly, enchytraeid communities of these peatlands are mainly dominated by *Cognettia sphagnetorum* (Carrera and Briones, 2013), a species complex (Martinsson et al., 2015) with specific

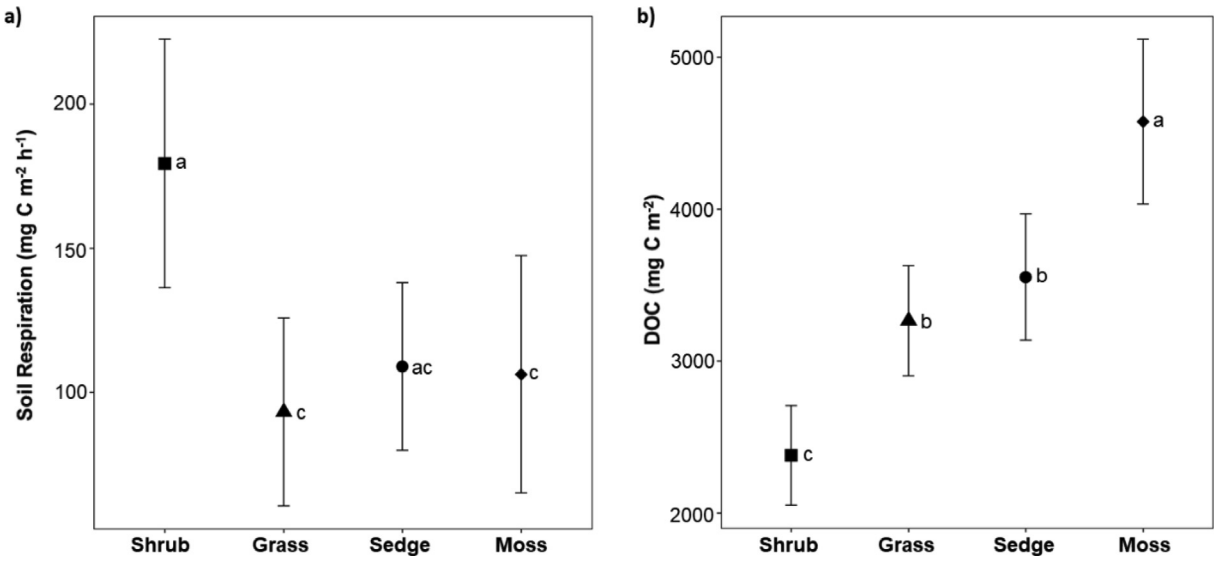


Fig. 2. Mean values (± 95% confidence intervals) of (a) soil respiration (n = 5 per habitat and sampling date) and (b) DOC (n = 3 per habitat and sampling date). Different letters indicate differences between habitats ($p < 0.05$).

ecophysiological adaptations to poor-nutrient environments. As such, *C. sphagnetorum* is stress-tolerant and asexual reproduction by fragmentation is the norm most of the time (Graefe and Schmelz, 1999).

In the three habitats dominated by vascular plants, mites were the most abundant soil mesofauna group and were mainly concentrated in the uppermost peat layers (Juan-Ovejero et al., 2019), where they can have access to a greater supply of labile substrates (Bonkowski et al., 2009). Additionally, DOC represents an easily assimilating carbon source for soil fauna living in the topsoil close to the litter surface (Liao et al., 2016).

The two habitats dominated by herbaceous plants (grass and sedge) released more DOC than the shrub habitat. Both grasses and sedges are fast growing species, that produce abundant litter, vast root systems and great amounts of root exudates containing labile compounds (Aerts, 1989; Moore et al., 2007; Basiliko et al., 2012) that can be rapidly mineralized or lost by leaching, especially with increasing temperatures (Dieleman et al., 2017; Leroy et al., 2017). In contrast, carbon release at the shrub habitat was mostly dominated by CO_2 emissions, which contradicts previous findings from a plant removal field experiment, which showed lower soil respiration rates in areas dominated by ericaceous shrubs than in areas dominated by other plant species (Ward et al., 2009). The discrepancy might be the result of a different experimental design, i.e. without removing plant species and with biotic and abiotic conditions co-varying over time. The presence of ericoid mycorrhizal fungi associated to the hair roots of ericaceous shrubs (Read and Perez-Moreno, 2003), together with the observed higher soil

Table 2
Selected Generalized Linear Mixed Models (GLMMs) for DOC and soil respiration at ecosystem level (variable addition and model selection performed through Type II Sum of Squares and $p < 0.05$ for Chi-Square Wald test).

	Explanatory variables: Abiotic factors				**Explanatory variables: Biotic factors**		
	Response variable: DOC				**Response variable: DOC**		
	Parameter value	**p-value**	**R^2(model)**		**Parameter value**	**p-value**	**R^2(model)**
Intercept	−2.524		0.394	Intercept	+0.320		0.206
	+0.030*Soil moisture	< 0.001			−0.349*Microbial biomass	< 0.001	
					−0.165*Acari biomass	< 0.001	
					Parameter value	**p-value**	**R^2(model)**
				Intercept	−0.115		0.172
					−0.128*Below-ground plant biomass	0.043	
					−0.171*Acari biomass	< 0.001	
					Parameter value	**p-value**	**R^2(model)**
				Intercept	+0.147		0.145
					+0.130*Below-ground plant biomass	0.042	
					−0.304*Microbial biomass	0.012	
					−0.113*Collembola biomass	0.017	
	Explanatory variables: Abiotic factors				**Explanatory variables: Biotic factors**		
	Response variable: Soil respiration				**Response variable: Soil respiration**		
	Parameter value	**p-value**	**R^2(model)**		**Parameter value**	**p-value**	**R^2(model)**
Intercept	+1.168		0.454	Intercept	−1.05		0.125
	−0.042*Soil moisture	0.002			+0.474*Microbial biomass	0.029	
	+0.111*Soil temperature	0.034			−0.199*Enchytraeidae biomass	< 0.001	
					Parameter value	**p-value**	**R^2(model)**
				Intercept	−0.252		0.175
					−0.468*Below-ground plant biomass	0.015	
					−0.173*Enchytraeidae biomass	< 0.001	

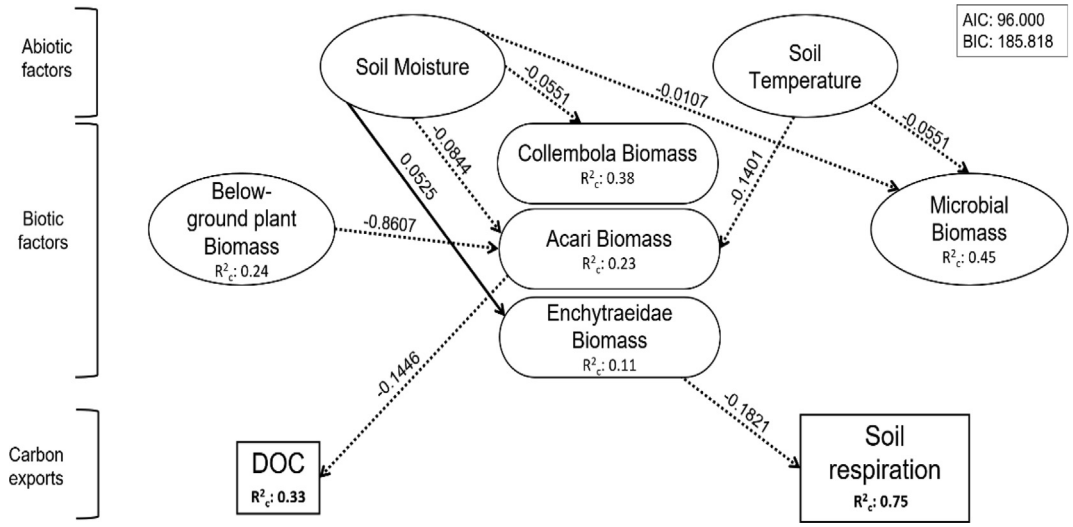


Fig. 3. Results of the piecewise Structural Equation Modelling (piecewise SEM) exploring the direct and indirect relationships between abiotic (soil moisture and soil temperature), biotic factors (Enchytraeidae biomass, Collembola biomass, Acari biomass, microbial biomass and below-ground plant biomass) and C release (soil respiration and DOC) at ecosystem level. Only significant paths are shown ($p < 0.05$). Uninterrupted arrows indicate positive relationships and dashed arrows negative relationships. The linear coefficients are shown above the arrows; $R^2_{conditional}$ of each endogenous variable is based on fixed and random effects.

temperatures and aerobic conditions may have benefited organic matter decomposition at the shrub habitat. In line with these findings, other studies have predicted that an increase in plant cover of this plant functional group could alter the carbon source capacity of peatlands (Bragazza et al., 2015; Kaštovská et al., 2018).

Moreover, the piecewise SEM indicated that below-ground plant biomass was the main factor affecting DOC release through its direct effect on Acari biomass. In agreement with this, below-ground plant biomass is one of the main sources of carbon for decomposer soil fauna, particularly for oribatid mites, as they can feed directly on living and dead roots and on fungi and bacteria living on root exudates (Pollierer et al., 2007). Therefore, below-ground plant biomass is a key biotic factor regulating carbon cycling in peatlands, and the relationship between primary plant production and DOC production has already been observed in wet peaty soils, where seasonal peaks in DOC release were significantly correlated to higher plant productivity (Harrison et al., 2008).

Enchytraeidae biomass was much higher than that of the two groups of microarthropods at the ecosystem level, and it was positively affected by soil moisture, which is in agreement with previous studies reporting that microarthropods are more sensitive than enchytraeids to high soil water contents (Bardgett, 2005; Holmstrup et al., 2012, 2013; Juan-Ovejero et al., 2019). Importantly, our SEM also showed that enchytraeids were negatively related to soil respiration, indicating that waterlogging conditions favoured enchytraied activities but suppressed CO_2 emissions at the ecosystem level.

Also at the ecosystem level, the amount of microbial biomass present in the peat was modulated by soil moisture and soil temperature, which is in agreement with other studies performed in peatlands (e.g. Mäkiranta et al., 2009; Delarue et al., 2015). Specifically, abiotic factors regulate the development of a hyphal network of soil fungi (Bonkowski et al., 2009; Eisenhauer et al., 2017), and other soil processes occurring in the soil that allow roots to transfer carbon to the soil, such as root exudation (Crow and Wieder, 2005; Keiser et al., 2019). The size of microbial biomass is commonly thought to be closely linked to the rate of carbon turnover in peatlands (Moore and Basiliko, 2006; Basiliko et al., 2007). However, reductions of microbial biomass caused by warming conditions and manipulations of mixotrophs abundance did not affect DOC availability in a peatland microcosm experiment (Jassey et al., 2015). In our piecewise SEM, microbial biomass did not show any relationship with carbon release, despite of being larger by several orders of magnitude than mesofaunal biomass.

5. Conclusions

Our well-replicated data obtained under natural field conditions allowed us to investigate the complex network of ecological relationships occurring at peatland ecosystem level, and to conclude that the abiotic factors, soil biota communities and carbon release differed depending on the dominant plant growth form (i.e. woody shrubs, herbaceous plants and non-vascular plants) growing at each habitat. We found that soil moisture and soil temperature had direct effects on the biological communities but affected carbon exports indirectly. By taking biomass as a proxy measure, we provide a better understanding of the role of various key groups of soil biota in carbon cycling in peatlands: while lower soil respiration rates were linked to enchytraeids, Acari had a more important influence in DOC production through interactions with below-ground plant biomass. These findings increase our knowledge on the intricate complex interactive activities occurring in the soil that could have a crucial effect on the carbon cycle (Osler and Sommerkorn, 2007; Briones, 2014; Filser et al., 2016), and therefore, provide a more holistic view of the soil processes controlling carbon exports in peatlands. Further research including functional traits of mesofauna could increase our understanding of the effects of soil organisms on carbon cycling in peatlands, since vertical stratification or phenology may affect carbon transformations (Fry et al., 2019).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank J. Pascual and J. Castro for their help with mesofauna identification, and M.E. Barreal and M.C.F. Lago for assistance with microbial biomass extractions. We also thank Bjorn Robroek and three anonymous reviewers for useful comments that improved the manuscript. R.J.-O. was supported by a PhD fellowship (FPI Programme, Ref. BES-2015-074461) and this work was funded by a MINECO research grant (Ref. CGL2014-54861-R).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geoderma.2020.114313.

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